

```
In [2]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [3]: df = pd.read_csv("mymoviedb.csv" , lineterminator = '\n')
```

```
In [4]: df.head()
```

```
Out[4]:
```

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Lan
--	--------------	-------	----------	------------	------------	--------------	--------------

0	2021-12-15	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	
---	------------	-------------------------	---	----------	------	-----	--

1	2022-03-01	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	
---	------------	------------	---	----------	------	-----	--

2	2022-02-25	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	
---	------------	---------	---	----------	-----	-----	--

3	2021-11-24	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	
---	------------	---------	---	----------	------	-----	--

4	2021-12-22	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	
---	------------	----------------	---	----------	------	-----	--



```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Release_Date          9827 non-null   object
1   Title                 9827 non-null   object
2   Overview              9827 non-null   object
3   Popularity            9827 non-null   float64
4   Vote_Count            9827 non-null   int64
5   Vote_Average          9827 non-null   float64
6   Original_Language     9827 non-null   object
7   Genre                 9827 non-null   object
8   Poster_Url           9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

```
In [14]: df['Genre'].head()
```

```
Out[14]: 0    Action, Adventure, Science Fiction
1          Crime, Mystery, Thriller
2                      Thriller
3    Animation, Comedy, Family, Fantasy
4    Action, Adventure, Thriller, War
Name: Genre, dtype: object
```

```
In [23]: df.duplicated().sum()
```

```
Out[23]: 0
```

```
In [24]: df.describe()
```

```
Out[24]:
```

	Popularity	Vote_Count	Vote_Average
count	9827.000000	9827.000000	9827.000000
mean	40.326088	1392.805536	6.439534
std	108.873998	2611.206907	1.129759
min	13.354000	0.000000	0.000000
25%	16.128500	146.000000	5.900000
50%	21.199000	444.000000	6.500000
75%	35.191500	1376.000000	7.100000
max	5083.954000	31077.000000	10.000000

Data Processing

```
In [31]: df['Release_Date'] = pd.to_datetime(df['Release_Date'])
print(df['Release_Date'].dtypes)
```

datetime64[ns]

```
In [32]: df['Release_Date'] = df['Release_Date'].dt.year
df['Release_Date'].dtypes
```

Out[32]: dtype('int32')

```
In [33]: df.head()
```

Out[33]:

	Release_Date	Title	Overview	Popularity	Vote_Count	Vote_Average	Original_Lan
0	2021	Spider-Man: No Way Home	Peter Parker is unmasked and no longer able to...	5083.954	8940	8.3	
1	2022	The Batman	In his second year of fighting crime, Batman u...	3827.658	1151	8.1	
2	2022	No Exit	Stranded at a rest stop in the mountains durin...	2618.087	122	6.3	
3	2021	Encanto	The tale of an extraordinary family, the Madri...	2402.201	5076	7.7	
4	2021	The King's Man	As a collection of history's worst tyrants and...	1895.511	1793	7.0	

```
In [34]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Release_Date          9827 non-null   int32
1   Title                 9827 non-null   object
2   Overview              9827 non-null   object
3   Popularity            9827 non-null   float64
4   Vote_Count            9827 non-null   int64
5   Vote_Average          9827 non-null   float64
6   Original_Language     9827 non-null   object
7   Genre                 9827 non-null   object
8   Poster_Url           9827 non-null   object
dtypes: float64(2), int32(1), int64(1), object(5)
memory usage: 652.7+ KB
```

Dropping the coloumn

```
In [50]: cols = ['Overview' , 'Poster_Url' , 'Original_Language']
```

```
In [58]: df.drop(cols, axis = 1, inplace = True)
df.columns
```

```
Out[58]: Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average',
               'Genre'],
              dtype='object')
```

```
In [59]: df.head()
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	8.3	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	8.1	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	6.3	Thriller
3	2021	Encanto	2402.201	5076	7.7	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	7.0	Action, Adventure, Thriller, War

catagorizing Vote_Average

```
In [91]: def categorize_col(df, col, labels):
edges = [df[col].describe()['min'],
df[col].describe()['25%'],
df[col].describe()['50%'],
df[col].describe()['75%'],
df[col].describe()['max']]
df[col] = pd.cut(df[col], edges, labels = labels, duplicates = 'drop')
return df
```

```
In [92]: labels = ['not_popular', 'below_average', 'average', 'popular']
categorize_col(df, 'Vote_Average', labels)
df['Vote_Average'].unique()
```

```
Out[92]: ['popular', 'below_average', 'average', 'not_popular', NaN]
Categories (4, object): ['not_popular' < 'below_average' < 'average' < 'popular']
```

```
In [93]: df.head()
```

```
Out[93]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	below_average	Thriller
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	average	Action, Adventure, Thriller, War

```
In [96]: df['Vote_Average'].value_counts()
```

```
Out[96]: Vote_Average
not_popular    2467
popular        2450
average        2412
below_average  2398
Name: count, dtype: int64
```

```
In [98]: df.dropna(inplace = True)
df.isna().sum()
```

```
Out[98]: Release_Date    0
         Title          0
         Popularity     0
         Vote_Count     0
         Vote_Average   0
         Genre          0
         dtype: int64
```

```
In [99]: df.head()
```

```
Out[99]:
```

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller
2	2022	No Exit	2618.087	122	below_average	Thriller
3	2021	Encanto	2402.201	5076	popular	Animation, Comedy, Family, Fantasy
4	2021	The King's Man	1895.511	1793	average	Action, Adventure, Thriller, War

Data Vizualization

```
In [ ]: sns.set_style('whitegrid')
```

What is the most frequwnt genre of movie released on Netflix

```
In [101... df['Genre'].describe()
```

```
Out[101... count    9727
unique    2324
top       Drama
freq      458
Name: Genre, dtype: object
```

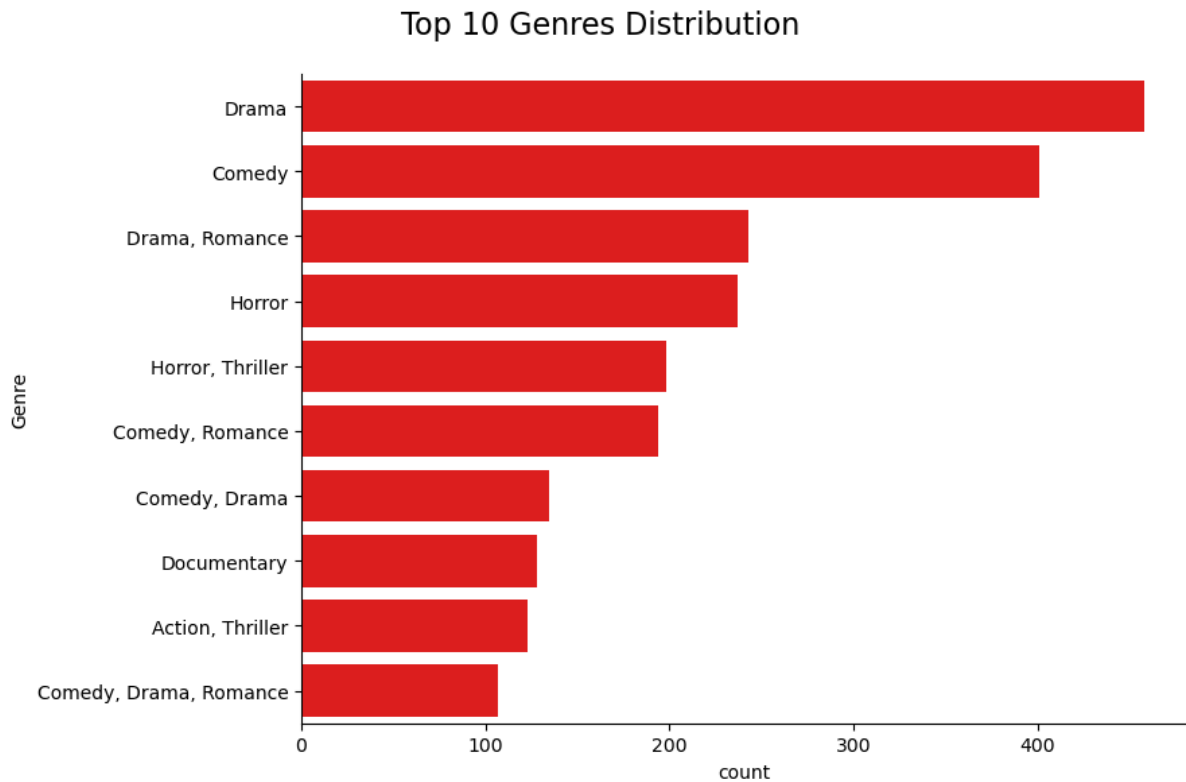
```
In [145... top_10_genres = df['Genre'].value_counts().head(10).index
df_top = df[df['Genre'].isin(top_10_genres)]
g = sns.catplot(
    y='Genre',
    data=df_top,
    kind='count',
    order=top_10_genres,
```

```

        color='red',
        height=6, aspect=1.5
    )
    g.fig.suptitle("Top 10 Genres Distribution", fontsize=16)
    g.fig.subplots_adjust(top=0.9)

plt.show()

```

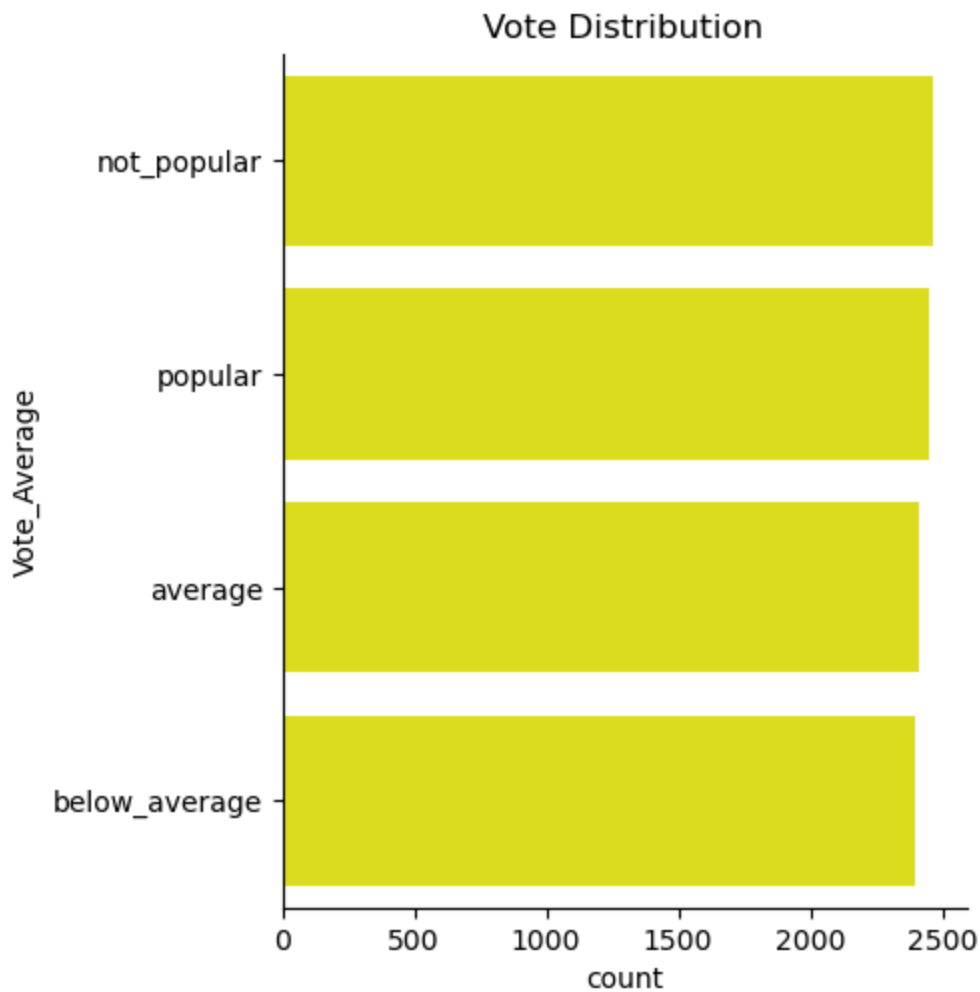


Which has highest votes in vote avg column

```

In [128... sns.catplot(y='Vote_Average', data=df, kind='count',
                order=df['Vote_Average'].value_counts().index,
                color='yellow')
plt.title("Vote Distribution")
plt.show()

```



what movie got the highest popularity?
What it's genre?

In [123...

df.head(2)

Out[123...

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction
1	2022	The Batman	3827.658	1151	popular	Crime, Mystery, Thriller

In [125...

df[df['Popularity'] == df['Popularity'].max()]

Out[125...

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
0	2021	Spider-Man: No Way Home	5083.954	8940	popular	Action, Adventure, Science Fiction

what movie got the lowest popularity?
what it's genre?

In [126...

```
df[df['Popularity'] == df['Popularity'].min()]
```

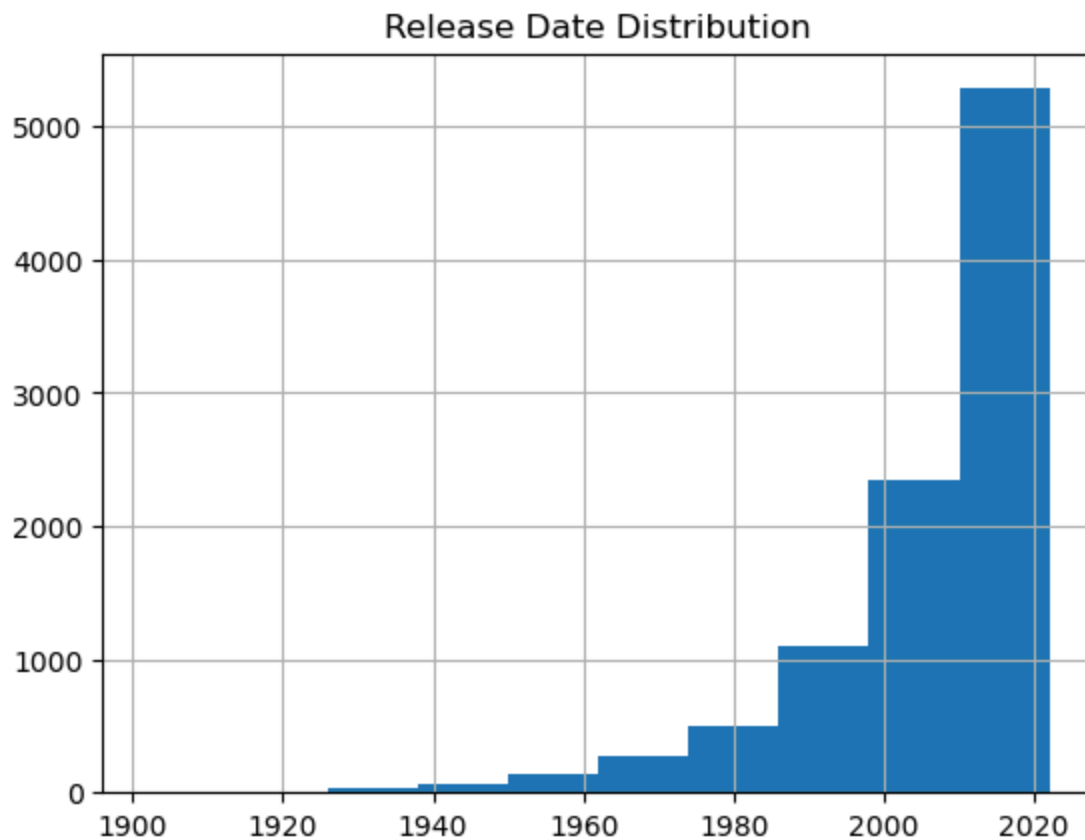
Out[126...

	Release_Date	Title	Popularity	Vote_Count	Vote_Average	Genre
9825	2021	The United States vs. Billie Holiday	13.354	152	average	Music, Drama, History
9826	1984	Threads	13.354	186	popular	War, Drama, Science Fiction

Which year has the most filmed movies?

In [131...

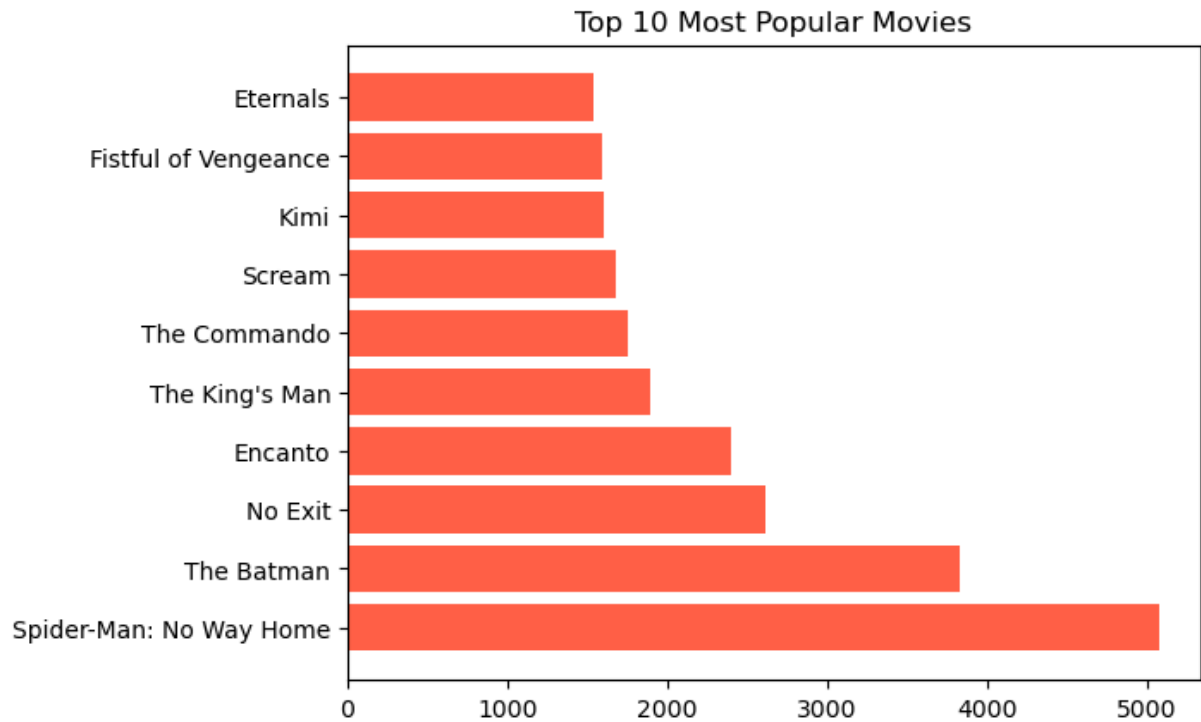
```
df['Release_Date'].hist()
plt.title("Release Date Distribution")
plt.show()
```



Top 10 Most Popular Movies

```
In [9]: top10 = df.nlargest(10, 'Popularity')

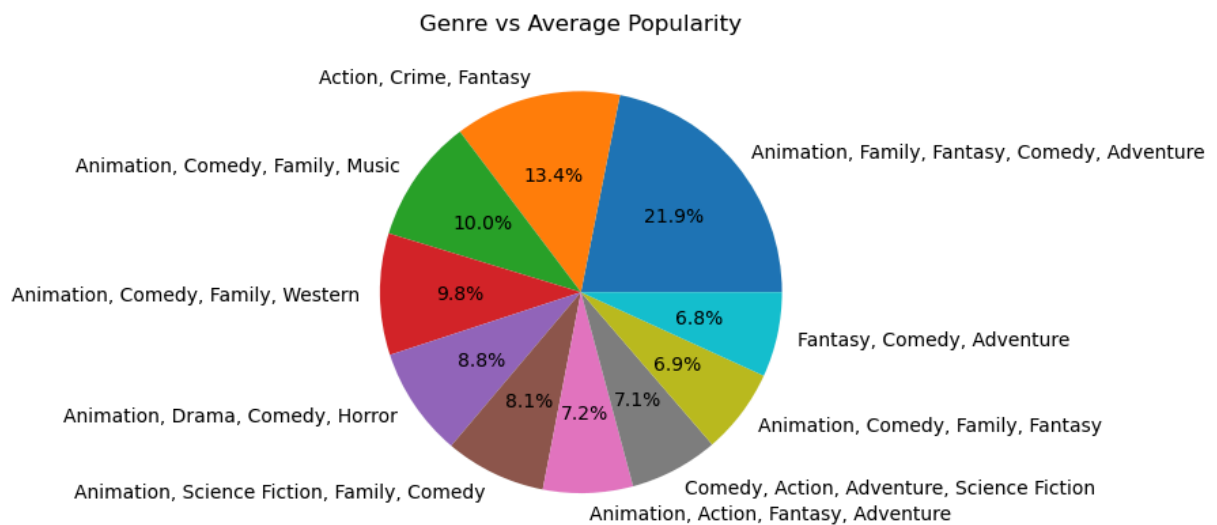
plt.barh(top10['Title'], top10['Popularity'], color='tomato')
plt.title("Top 10 Most Popular Movies")
plt.show()
```



Genre vs Average Popularity

```
In [10]: genre_pop = df.groupby('Genre')['Popularity'].mean().nlargest(10)

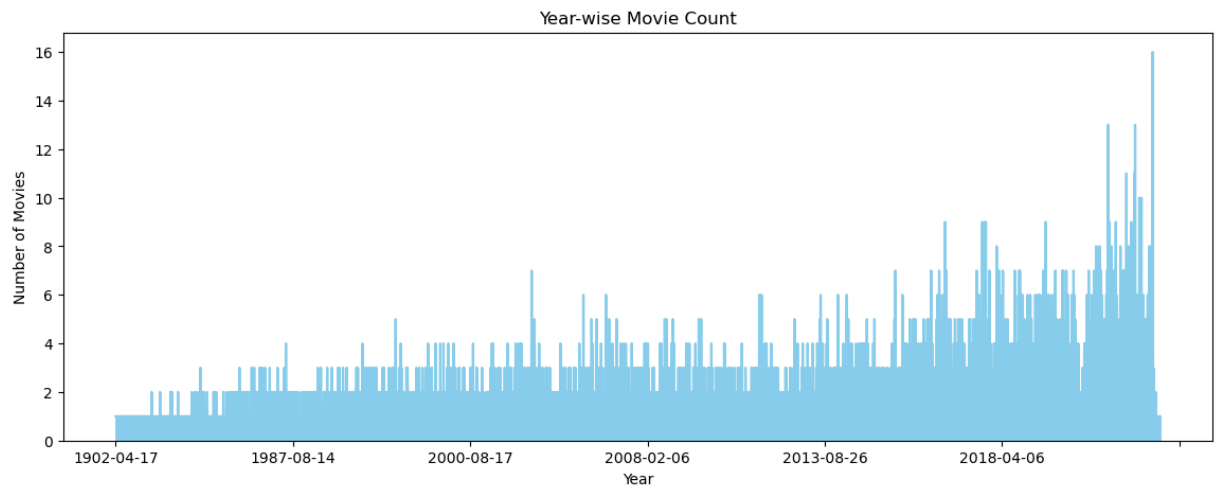
genre_pop.plot(kind='pie', autopct='%1.1f%%')
plt.title("Genre vs Average Popularity")
plt.ylabel("")
plt.show()
```



Year-wise Movie Release Trend

```
In [16]: yearly = df['Release_Date'].value_counts().sort_index()

plt.figure(figsize=(14, 5))
yearly.plot(kind='area', color='skyblue')
plt.title("Year-wise Movie Count")
plt.xlabel("Year")
plt.ylabel("Number of Movies")
plt.show()
```



```
In [ ]:
```